STAT 559 – Measure Theory
Collected Edited Lecture Notes

English transcription with corrections and supplements

These notes transcribe and edit the indicated handwritten PDF. Source-page markers preserve traceability. Yellow boxes retain original TODOs; red boxes explain mathematical corrections; blue boxes add explanations or proof details. The reference is *stats310notes.pdf* (provided in `notes/`); references below use its printed section numbers.

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Part I

STAT 559 – Note 01: Measure Spaces and Extension: Foundations

Source page(s): 1–3

1 Why measure theory?

The central question is how to define length on $\mathbb{R}$, area on $\mathbb{R}^2$, and volume on $\mathbb{R}^3$, especially for complicated sets. The notes connect Borel's measurable spaces, set theory, and probability. Let Ω be a set with elements $\omega \in \Omega$. A *sigma-algebra* (or sigma-field) $\mathcal{F} \subseteq 2^\Omega$ satisfies

$$\varnothing \in \mathcal{F}, \tag{1}$$

$$A \in \mathcal{F} \Longrightarrow A^c = \Omega \setminus A \in \mathcal{F}, \tag{2}$$

$$A_1, A_2, \ldots \in \mathcal{F} \Longrightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}. \tag{3}$$

The pair $(\Omega, \mathcal{F})$ is a *measurable space*. A *measure* on it is a function $\mu : \mathcal{F} \to [0, \infty]$ such that $\mu(\varnothing) = 0$ and, for pairwise disjoint measurable sets,

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n). \tag{4}$$

The triple $(\Omega, \mathcal{F}, \mu)$ is a *measure space*.

Original TODO: Why restrict the sets that can be measured?

The handwritten illustration mentions the Banach–Tarski ("sun–pea") paradox: finitely many pieces of a three-dimensional ball can be rearranged by rigid motions into two balls of the original size. The pieces cannot all be measurable for the usual volume. This motivates choosing a suitable sigma-algebra rather than requiring every subset to have geometric volume. The paradox is motivational here, not proved in the supplied reference.

Correction: Degenerate measures and geometric volume

The source suggests allowing only zero or infinity as an alternative. The rule $\mu(\varnothing) = 0$ and $\mu(A) = \infty$ for every nonempty A is indeed a measure, but it does not extend ordinary length or volume. A useful geometric measure must retain the prescribed values on intervals or boxes.

Source page(s): 4–6

2 Generated sigma-algebras and topology

A sigma-algebra contains $\Omega = \varnothing^c$ and is closed under countable intersections, by De Morgan's law. The intersection of any nonempty family of sigma-algebras on the same Ω is again a sigma-algebra. For any class $\mathcal{A} \subseteq 2^\Omega$, the *sigma-algebra generated by* $\mathcal{A}$ is

$$\sigma(\mathcal{A}) = \bigcap \{\mathcal{G} : \mathcal{G} \text{ is a sigma-algebra on } \Omega,\ \mathcal{A} \subseteq \mathcal{G}\}. \tag{5}$$

The family being intersected is nonempty because it contains 2^Ω. The intersection is a sigma-algebra, contains $\mathcal{A}$, and is contained in every sigma-algebra containing $\mathcal{A}$. This proves existence and minimality. A *topology* $\tau \subseteq 2^\Omega$ contains $\varnothing, \Omega$ and is closed under arbitrary unions and finite intersections. Its members are open sets. The Borel sigma-algebra is $\mathcal{B}(\Omega) = \sigma(\tau)$.

Correction: Collections versus elements

The source calls a topology a subset of Ω. It is a collection of subsets, hence a subset of 2^Ω. Likewise, the family of sigma-algebras in the preceding proof is an indexed family; it need not be countably enumerable.

Source page(s): 7–8

On $\mathbb{R}$ with its usual topology, $\mathcal{B}(\mathbb{R})$ is generated by all open subsets. Measurable sets in a probability space are called *events*. A probability space is a measure space with $\mu(\Omega) = 1$.

3 Basic properties of measures

If $A \subseteq B$ are measurable, disjoint additivity gives

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A). \tag{6}$$

For arbitrary measurable A_n, the *union bound* is

$$\mu\left(\bigcup_{n\geq 1} A_n\right) \leq \sum_{n\geq 1} \mu(A_n). \tag{7}$$

For example, disjointify the union using $D_1 = A_1$ and $D_n = A_n \setminus \bigcup_{j<n} A_j$; then apply additivity and $D_n \subseteq A_n$. If $A_n \uparrow A$, then

$$\mu(A) = \lim_{n\to\infty} \mu(A_n). \tag{8}$$

Indeed, apply countable additivity to $A_1, A_2 \setminus A_1, \ldots$. If $A_n \downarrow A$ and $\mu(A_1) < \infty$, then

$$\mu(A) = \lim_{n\to\infty} \mu(A_n). \tag{9}$$

Apply continuity from below to $A_1 \setminus A_n$ and subtract from $\mu(A_1)$.

Original TODO: Why is finiteness needed for continuity from above?

The source assumes the stronger condition $\mu(\Omega) < \infty$. It suffices that one set of the decreasing sequence has finite measure. Without this, take Lebesgue measure and $A_n = [n, \infty)$: every $\mu(A_n) = \infty$, while $\bigcap_n A_n = \varnothing$ has measure zero. See the reference, Section 1.2, continuity of measures.

Source page(s): 9–10

4 Algebras and approximation

An *algebra* $\mathcal{A}$ contains $\varnothing$, is closed under complements, and is closed under finite unions.

Correction: Finite union in the definition

The source writes an infinite upper limit next to the words "finite union." The intended condition is $\bigcup_{j=1}^{n} A_j \in \mathcal{A}$ for every finite n.

Approximation theorem. Suppose $\mu(\Omega) < \infty$ and $\mathcal{F} = \sigma(\mathcal{A})$, where $\mathcal{A}$ is an algebra. For every $A \in \mathcal{F}$ and $\varepsilon > 0$ there is $B \in \mathcal{A}$ with

$$\mu(A \triangle B) < \varepsilon, \qquad A \triangle B = (A \setminus B) \cup (B \setminus A). \tag{10}$$

The notes state this for a probability space. Define $\mathcal{G}$ as the measurable sets with this approximation property. Then $\mathcal{A} \subseteq \mathcal{G} \subseteq \mathcal{F}$. Complements preserve symmetric differences, so $\mathcal{G}$ is closed under complements. For $A_n \in \mathcal{G}$, put $A = \bigcup_n A_n$ and $C_N = \bigcup_{n \le N} A_n$. By continuity and finiteness, choose N with $\mu(A \setminus C_N) < \varepsilon/2$. Choose $B_n \in \mathcal{A}$ for $n \le N$ with $\mu(A_n \triangle B_n) < \varepsilon/(2N)$. Then $B = \bigcup_{n \le N} B_n \in \mathcal{A}$ and

$$\mu(A \triangle B) \le \mu(A \setminus C_N) + \sum_{n=1}^{N} \mu(A_n \triangle B_n) < \varepsilon. \tag{11}$$

Thus $\mathcal{G}$ is a sigma-algebra containing $\mathcal{A}$, so $\mathcal{G} = \mathcal{F}$.

Supplement: Completing the proof sketch

The handwritten proof gives the sigma-algebra strategy but leaves its closure under countable unions to be checked. The finite truncation argument above supplies that step. Compare the reference Theorem 1.2.6.

Source page(s): 11–12

5 Constructing length and extending a measure

The desired interval values are $\mu((a, b]) = b - a$ for real $a \le b$. Begin with simple intervals, prescribe their lengths, and extend to their generated sigma-algebra.

Correction: The starting class and uniqueness hypotheses

The source alternates between open and half-open intervals, and calls the initial triple a measure space before a sigma-algebra has been constructed. A consistent construction starts with half-open intervals (and the empty set), extends length to their finite disjoint unions, and obtains a premeasure on that algebra. Infinite endpoints need appropriate ray conventions. Carathéodory extension then gives a measure on the generated Borel sigma-algebra. Uniqueness requires an appropriate sigma-finiteness assumption; it is not automatic for an arbitrary initial set function. Its completion is Lebesgue measure on the Lebesgue sigma-algebra. See reference Sections 1.5–1.7.

Source page(s): 13–14

6 The pi–lambda theorem and uniqueness

A *pi-system* $\mathcal{P}$ is a class closed under finite nonempty intersections. A *lambda-system* $\mathcal{L}$ contains Ω, is closed under complements, and is closed under countable disjoint unions. A lambda-system that is also a pi-system is a sigma-algebra: intersections and complements give finite unions and differences; disjointification then gives countable unions. **Dynkin's pi–lambda theorem.** If $\mathcal{P}$ is a pi-system and $\mathcal{L}$ is a lambda-system with $\mathcal{P} \subseteq \mathcal{L}$, then $\sigma(\mathcal{P}) \subseteq \mathcal{L}$. **Uniqueness theorem.** Let μ_1, μ_2 be measures on $\sigma(\mathcal{P})$ agreeing on a pi-system $\mathcal{P}$. Suppose there are $E_n \in \mathcal{P}$ with $E_n \uparrow \Omega$ and $\mu_1(E_n) = \mu_2(E_n) < \infty$. Then $\mu_1 = \mu_2$ on $\sigma(\mathcal{P})$.

Source page(s): 15–16

Proof. Fix $E \in \mathcal{P}$ of finite common measure, and set

$$\mathcal{L}_E = \{B \in \sigma(\mathcal{P}) : \mu_1(E \cap B) = \mu_2(E \cap B)\}. \tag{12}$$

It contains Ω. If $B \in \mathcal{L}_E$, finite subtraction gives

$$\begin{aligned} \mu_1(E \cap B^c) &= \mu_1(E) - \mu_1(E \cap B) && (13)\\ &= \mu_2(E) - \mu_2(E \cap B) = \mu_2(E \cap B^c). && (14) \end{aligned}$$

For disjoint $B_n \in \mathcal{L}_E$,

$$\begin{aligned} \mu_1\left(E \cap \bigcup_n B_n\right) &= \sum_n \mu_1(E \cap B_n) && (15)\\ &= \sum_n \mu_2(E \cap B_n) = \mu_2\left(E \cap \bigcup_n B_n\right). && (16) \end{aligned}$$

Thus $\mathcal{L}_E$ is a lambda-system. Since $E \cap B \in \mathcal{P}$ for $B \in \mathcal{P}$, it contains $\mathcal{P}$. Dynkin's theorem gives $\mathcal{L}_E = \sigma(\mathcal{P})$. Therefore, for all $B \in \sigma(\mathcal{P})$, continuity from below yields

$$\mu_1(B) = \lim_n \mu_1(E_n \cap B) = \lim_n \mu_2(E_n \cap B) = \mu_2(B). \tag{17}$$

Source page(s): 17

7 Carathéodory's roadmap

First extend a premeasure from an algebra to an outer measure on all subsets. Next restrict that outer measure to its Carathéodory-measurable sets, which form a sigma-algebra. Finally restrict to the sigma-algebra generated by the original algebra:

$$(\mathcal{A}, \mu_0) \longrightarrow (2^\Omega, \mu^*) \longrightarrow (\mathcal{M}_{\mu^*}, \mu^*|_{\mathcal{M}_{\mu^*}}) \longrightarrow (\sigma(\mathcal{A}), \mu^*|_{\sigma(\mathcal{A})}). \tag{18}$$

The first object in the middle carries an outer measure, not in general a measure. The next note supplies the two Carathéodory theorems indicated in the source diagram.

Part II

STAT 559 – Note 02: Outer Measures, Extension, and Completion

Source page(s): 1–2

8 Outer measures and their measurable sets

The construction has three stages: extend to all subsets as an outer measure, restrict to the sets measurable for that outer measure, then restrict further to the desired generated sigma-algebra. An *outer measure* is $\phi : 2^\Omega \to [0, \infty]$ such that

$$\phi(\varnothing) = 0, \tag{19}$$

$$A \subseteq B \Longrightarrow \phi(A) \leq \phi(B), \tag{20}$$

$$\phi\left(\bigcup_{n\geq 1} A_n\right) \leq \sum_{n\geq 1} \phi(A_n). \tag{21}$$

Supplement: The source's question: is the power set a sigma-algebra?

Yes, 2^Ω is a sigma-algebra. An outer measure need not be a measure on it because countable *additivity* on disjoint sets is not assumed. Countable subadditivity alone does not give the volume contradiction discussed with Banach–Tarski. Outer measure is defined on all subsets; the power set is not assumed to be a proper subset of anything.

A set $A \subseteq \Omega$ is *Carathéodory-measurable for* ϕ if for every $E \subseteq \Omega$,

$$\phi(E) = \phi(E \cap A) + \phi(E \cap A^c). \tag{22}$$

Let $\mathcal{M}_\phi$ be the class of these sets. The central theorem is that $\mathcal{M}_\phi$ is a sigma-algebra and $\phi|_{\mathcal{M}_\phi}$ is a measure (reference Theorem 1.4.3).

Source page(s): 3–4

9 Proof: algebra and finite additivity

The empty set is measurable because $\phi(E) = 0 + \phi(E)$. The defining equality is symmetric in A, A^c, proving complement closure. For $A, B \in \mathcal{M}_\phi$, put $D = A \cup B$. Then

$$\phi(E \cap D) + \phi(E \cap D^c) \leq \phi(E \cap A) + \phi(E \cap A^c \cap B) + \phi(E \cap A^c \cap B^c) \tag{23}$$

$$= \phi(E \cap A) + \phi(E \cap A^c) = \phi(E). \tag{24}$$

The opposite inequality is outer subadditivity. Thus $D \in \mathcal{M}_\phi$ and $\mathcal{M}_\phi$ is an algebra. For disjoint $A_1, \ldots, A_n \in \mathcal{M}_\phi$ and any $E \subseteq \Omega$,

$$\phi\left(E \cap \bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} \phi(E \cap A_i). \tag{25}$$

Source page(s): 5–6

Write $C_n = \bigcup_{i=1}^{n} A_i$. Splitting $E \cap C_n$ by measurable A_n gives

$$\phi(E \cap C_n) = \phi(E \cap A_n) + \phi(E \cap C_{n-1}), \tag{26}$$

which proves the identity by induction. If measurable $A_n \uparrow A$, the measurability of A is not yet known. Set $D_1 = A_1$ and $D_n = A_n \setminus A_{n-1}$ for $n \geq 2$. They are disjoint members of the algebra, and

$$\phi(E \cap A_n) = \sum_{i=1}^{n} \phi(E \cap D_i), \tag{27}$$

$$\lim_n \phi(E \cap A_n) = \sum_{i=1}^{\infty} \phi(E \cap D_i) \geq \phi(E \cap A). \tag{28}$$

Supplement: Passing from finite to infinite sums

The source asks how the limit is justified. For nonnegative summands the infinite series is, by definition, the increasing limit of its partial sums. The last inequality is countable subadditivity, not an application of measure continuity before the theorem has been proved.

Source page(s): 6–8

10 Proof: sigma-algebra and countable additivity

For arbitrary $A_i \in \mathcal{M}_\phi$, put $A = \bigcup_i A_i$ and $C_n = \bigcup_{i \leq n} A_i$. Since C_n is measurable and $A^c \subseteq C_n^c$,

$$\phi(E) = \phi(E \cap C_n) + \phi(E \cap C_n^c) \geq \phi(E \cap C_n) + \phi(E \cap A^c). \tag{29}$$

Taking the increasing limit and using the previous lemma gives

$$\phi(E) \geq \phi(E \cap A) + \phi(E \cap A^c). \tag{30}$$

Subadditivity gives equality. Thus $A \in \mathcal{M}_\phi$. For disjoint measurable A_i, monotonicity and finite additivity give

$$\phi\left(\bigcup_i A_i\right) \geq \sum_{i=1}^{n} \phi(A_i) \quad \text{for every } n. \tag{31}$$

Let $n \to \infty$ and combine with subadditivity to obtain countable additivity. The source also recalls the separate uniqueness step, proved by the pi–lambda theorem under a common sigma-finite cover.

Source page(s): 9–10

11 Premeasures and extension

A *premeasure* on an algebra $\mathcal{A}$ is a function $\mu_0 : \mathcal{A} \to [0, \infty]$ with $\mu_0(\varnothing) = 0$ and

$$\mu_0\left(\bigcup_i A_i\right) = \sum_i \mu_0(A_i) \tag{32}$$

whenever $A_i \in \mathcal{A}$ are disjoint and their union also belongs to $\mathcal{A}$. **Carathéodory extension theorem** (reference Theorem 1.5.1). A premeasure on an algebra extends to a measure on $\sigma(\mathcal{A})$. If $\Omega = \bigcup_n E_n$ for $E_n \in \mathcal{A}$ with $\mu_0(E_n) < \infty$, the extension is unique. Define, for arbitrary $S \subseteq \Omega$,

$$\mu^*(S) = \inf\left\{\sum_{i=1}^{\infty} \mu_0(C_i) : C_i \in \mathcal{A},\ S \subseteq \bigcup_{i=1}^{\infty} C_i\right\}. \tag{33}$$

The collection of covers is nonempty, since $\Omega, \varnothing, \ldots$ is available; the infimum can equal infinity.

Source page(s): 11–12

12 The covering functional is an outer measure

The empty cover by empty sets proves $\mu^*(\varnothing) = 0$. Every cover of T covers $S \subseteq T$, so $\mu^*(S) \le \mu^*(T)$. To prove subadditivity, take arbitrary S_i. If $\sum_i \mu^*(S_i) = \infty$, the inequality is immediate. Otherwise, for $\varepsilon > 0$ choose covers $S_i \subseteq \bigcup_j C_{ij}$ with

$$\sum_j \mu_0(C_{ij}) < \mu^*(S_i) + 2^{-i}\varepsilon. \tag{34}$$

Their union is a countable cover of $\bigcup_i S_i$, hence

$$\mu^*\left(\bigcup_i S_i\right) \le \sum_{i,j} \mu_0(C_{ij}) \tag{35}$$

$$\le \sum_i \mu^*(S_i) + \varepsilon \sum_{i=1}^{\infty} 2^{-i} = \sum_i \mu^*(S_i) + \varepsilon. \tag{36}$$

Let $\varepsilon \downarrow 0$.

Original TODO: Check the definition of the infimum

The source marks the near-minimizing-cover step. For a finite infimum, some admissible cover has cost below the infimum plus any positive tolerance; otherwise that larger number would still be a lower bound. The choice $2^{-i}\varepsilon$ makes the total error at most ε. Nonnegative double sums may be enumerated in any order. The separate infinite case above avoids treating infinity as a finite approximation target. See reference Lemma 1.5.2.

Source page(s): 11, 13

Supplement: The homework step: agreement on the algebra

For $A \in \mathcal{A}$ the cover $A, \varnothing, \ldots$ gives $\mu^*(A) \le \mu_0(A)$. For any cover $A \subseteq \bigcup_i C_i$, disjointify $A \cap C_i$ within A. The resulting disjoint algebra sets D_i have union A, so

$$\mu_0(A) = \sum_i \mu_0(D_i) \le \sum_i \mu_0(C_i). \tag{37}$$

Taking the infimum gives $\mu_0(A) \le \mu^*(A)$. Thus equality holds (reference Lemma 1.5.3).

Source page(s): 13–14

13 Every algebra set is outer-measurable

Fix $A \in \mathcal{A}$, any $E \subseteq \Omega$, and an algebra cover $E \subseteq \bigcup_i C_i$. The sets $C_i \cap A$ cover $E \cap A$, and $C_i \cap A^c$ cover $E \cap A^c$. Finite additivity on $\mathcal{A}$ gives

$$\mu^*(E \cap A) + \mu^*(E \cap A^c) \leq \sum_i [\mu_0(C_i \cap A) + \mu_0(C_i \cap A^c)] \tag{38}$$

$$= \sum_i \mu_0(C_i). \tag{39}$$

Take the infimum over covers. The reverse inequality is subadditivity, so $A \in \mathcal{M}_{\mu^*}$. Consequently $\sigma(\mathcal{A}) \subseteq \mathcal{M}_{\mu^*}$, and restriction gives the required measure extension.

Correction: The infimum is a lower envelope

The source's page 14 contains an intermediate inequality suggesting a cover cost is at most $\mu^*(E)$. Every cover cost is instead at least the infimum. The argument works because the same left side is bounded by *every* cover cost; taking their infimum gives the desired bound.

Source page(s): 15–16

14 Lebesgue measure and completion

Let $\mathcal{A}$ be the algebra of finite disjoint unions of half-open intervals $(a, b] \cap \mathbb{R}$, allowing extended endpoints with the natural ray conventions. Assign each bounded interval its length $b - a$ and each unbounded nonempty interval infinite length; sum over disjoint components. The source assigns as homework proving that $\mathcal{A}$ is an algebra and this functional is a premeasure. Reference Section 1.6 supplies this construction. Since $\sigma(\mathcal{A}) = \mathcal{B}(\mathbb{R})$ and $\mathbb{R} = \bigcup_n (-n, n]$, the Borel extension is unique. Its completed version is the usual complete Lebesgue measure. The construction extends to $\mathbb{R}^n$; product measures are discussed later. A measure space is *complete* if every subset of every measurable null set is measurable.

Original TODO: A counterexample to automatic completeness

For a small explicit example, let $\Omega = \{a, b, c\}$ and $\mathcal{F} = \{\varnothing, \{a, b\}, \{c\}, \Omega\}$. Put mass zero on $\{a, b\}$ and mass one on $\{c\}$. Then $\{a\}$ is a subset of a measurable null set but is not in $\mathcal{F}$.

Supplement: The source's Borel-measurability question

Borel Lebesgue measure on $\mathcal{B}(\mathbb{R})$ is not complete. The Cantor set is Borel, null, and has cardinality $|\mathbb{R}|$. Its power set has strictly greater cardinality than the class of Borel sets, so some subsets of it are not Borel. They are measurable in the completion. This cardinality argument explains the example; the completion theorem itself is reference Theorem 1.7.2.

Source page(s): 17

15 Completing a measure space

For any measure space $(\Omega, \mathcal{F}, \mu)$, define

$$\overline{\mathcal{F}} = \{A \cup N : A \in \mathcal{F},\ N \subseteq Z \text{ for some } Z \in \mathcal{F},\ \mu(Z) = 0\}, \tag{40}$$

$$\overline{\mu}(A \cup N) = \mu(A). \tag{41}$$

Supplement: The completion homework

The formula is independent of the representation: two measurable parts differ only inside a measurable null set, so their measures agree. Complements and countable unions remain in this class by absorbing the exceptional subsets into countable unions of measurable null sets. Countable additivity follows by removing those null exceptional sets and using additivity on $\mathcal{F}$. Every subset of an $\overline{\mu}$-null measurable set is again of the indicated form, so this measure space is complete. Alternatively apply the covering construction to $\mathcal{F}$ itself and restrict to its Carathéodory-measurable sets, as proposed in the source. Every outer-null set is Carathéodory-measurable, which proves completeness of that resulting space (reference Section 1.7).

Part III

STAT 559 – Note 03: Measurable Functions and Lebesgue Integration

Source page(s): 1

16 Motivation

Lebesgue integration starts with a measure and builds integration from it. On a general space we write $\int_\Omega f(\omega)\,d\mu(\omega)$. With Lebesgue measure on a real interval this agrees with the Riemann integral whenever the bounded function is Riemann integrable (reference Exercise 2.4.2). It applies to many more measurable functions and supports powerful limit theorems.

Correction: Existence of an integral

Measurability alone does not guarantee a finite integral, or even a defined signed integral. A nonnegative measurable function always has an extended integral in $[0,\infty]$. A signed integral must avoid the indeterminate expression infinity minus infinity.

Source page(s): 2–3

17 Measurable functions

For measurable spaces $(\Omega,\mathcal{F})$ and $(\Omega',\mathcal{F}')$, a map $f:\Omega\to\Omega'$ is measurable if $f^{-1}(A)\in\mathcal{F}$ for every $A\in\mathcal{F}'$. **Generator criterion** (reference Lemma 2.1.3). If $\mathcal{F}'=\sigma(\mathcal{A})$ and $f^{-1}(A)\in\mathcal{F}$ for all $A\in\mathcal{A}$, then f is measurable. Indeed,

$$\mathcal{G}=\{B\subseteq\Omega' : f^{-1}(B)\in\mathcal{F}\} \tag{42}$$

is a sigma-algebra because inverse images preserve complements and countable unions. It contains $\mathcal{A}$, hence $\mathcal{F}'$. Continuous maps between topological spaces are Borel measurable: the inverse image of an open set is open, and open sets generate the Borel sigma-algebra. Compositions of measurable maps are measurable, since $(g\circ f)^{-1}(A)=f^{-1}(g^{-1}(A))$. For real-valued measurable f,g, the functions $f+g, f-g, fg$ are measurable. The quotient f/g is measurable wherever $g\neq 0$, with a specified measurable extension on its zero set if needed.

Source page(s): 4–5

18 Extended real functions and limits

Write $\overline{\mathbb{R}}=\mathbb{R}\cup\{-\infty,\infty\}$, with its order topology and Borel sigma-algebra. Countable infima and suprema of measurable extended real functions are measurable. For example,

$$\Big\{\inf_n f_n\geq t\Big\}=\bigcap_n\{f_n\geq t\}. \tag{43}$$

The upper rays $[t, \infty]$ for finite real t generate $\mathcal{B}(\overline{\mathbb{R}})$. The supremum result follows by negation. Thus

$$\liminf_n f_n = \sup_k \inf_{n \geq k} f_n, \tag{44}$$

$$\limsup_n f_n = \inf_k \sup_{n \geq k} f_n \tag{45}$$

are measurable, as is any everywhere pointwise limit.

Supplement: The source's question about open sets

The open sets in $\mathbb{R}$ and $\overline{\mathbb{R}}$ are not literally the same: neighborhoods of the added endpoints include sets such as $(a, \infty]$. On $\mathbb{R}$ an open set is a countable union of open intervals, but an arbitrary open set need not be a single interval. Its usual topology is the subspace topology inherited from $\overline{\mathbb{R}}$.

Source page(s): 6–8

19 Four stages in defining the integral

Fix a measure space $(\Omega, \mathcal{F}, \mu)$. Measurability of f refers to the domain sigma-algebra and target Borel sigma-algebra; the domain measure supplies the integral. First, for $A \in \mathcal{F}$, set

$$\int \mathbf{1}_A \, d\mu = \mu(A). \tag{46}$$

Second, a *nonnegative simple function* has a finite representation $s = \sum_{i=1}^m a_i \mathbf{1}_{A_i}$ with finite $a_i \geq 0$ and disjoint measurable A_i. Define

$$\int s \, d\mu = \sum_{i=1}^m a_i \mu(A_i), \qquad 0 \cdot \infty := 0. \tag{47}$$

Third, for measurable $f \geq 0$, let $\mathcal{S}_f$ be the nonnegative simple functions bounded above by f. Define

$$\int f \, d\mu = \sup_{s \in \mathcal{S}_f} \int s \, d\mu. \tag{48}$$

Original TODO: Prove linearity for simple functions

Refine two finite simple representations by the measurable intersections of their level sets (including zero level sets to partition the domain). On each common cell both functions are constant. Distribute the finite sums to prove

$$\int (as + bt) \, d\mu = a \int s \, d\mu + b \int t \, d\mu \quad (a, b \geq 0). \tag{49}$$

The same refinement proves that the integral is independent of the chosen representation. See reference Section 2.2.

Source page(s): 9–10

Fourth, put $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$. Then $f = f^+ - f^-$ and $|f| = f^+ + f^-$. Define

$$\int f \, d\mu = \int f^+ \, d\mu - \int f^- \, d\mu \tag{50}$$

if at least one term is finite. The function is *integrable* when both are finite, equivalently $\int |f| \, d\mu < \infty$. For measurable S, $\int_S f \, d\mu = \int f \mathbf{1}_S \, d\mu$, interpreting the product to be zero outside S even if f is infinite there.

Original TODO: An undefined signed integral

On $\mathbb{R}$ with Lebesgue measure, let $f(x) = 1$ for $x \geq 0$ and $f(x) = -1$ for $x < 0$. Both $\int f^+ \, d\lambda$ and $\int f^- \, d\lambda$ are infinite, so its signed Lebesgue integral is undefined.

Source page(s): 11–12

20 Monotone convergence and simple approximation

Monotone convergence theorem (MCT). For measurable $0 \leq f_n \uparrow f$ pointwise,

$$\lim_n \int f_n \, d\mu = \int f \, d\mu. \tag{51}$$

The limit f is measurable. A key companion fact is that every measurable $f \geq 0$ is an increasing limit of nonnegative simple functions (reference Proposition 2.3.6).

Supplement: The construction assigned as homework

One explicit approximation is

$$s_n = 2^{-n} \lfloor 2^n \min(f, n) \rfloor . \tag{52}$$

For $f = \infty$ take $\min(f, n) = n$. Each s_n has finitely many measurable level sets; refinement of the dyadic grid and increasing truncation show $s_n \leq s_{n+1} \leq f$. For finite $f(\omega)$, the approximation error is eventually less than 2^{-n}; at infinity, $s_n(\omega) = n \to \infty$. Thus $s_n \uparrow f$.

Source page(s): 13–14

21 Linearity

For measurable nonnegative f, g, choose $u_n \uparrow f$ and $v_n \uparrow g$ simple. For $a, b \geq 0$, MCT and simple-function linearity give

$$\int (af + bg) \, d\mu = \lim_n \int (au_n + bv_n) \, d\mu \tag{53}$$

$$= a \lim_n \int u_n \, d\mu + b \lim_n \int v_n \, d\mu \tag{54}$$

$$= a \int f \, d\mu + b \int g \, d\mu. \tag{55}$$

Original TODO: Complete the signed linearity proof

For integrable f, g, their infinite values occur only on a measurable null set. Define ambiguous sums there to be zero. The inequalities $(f+g)^{\pm} \le f^{\pm} + g^{\pm}$ show integrability. The identity

$$(f+g)^{+} + f^{-} + g^{-} = (f+g)^{-} + f^{+} + g^{+} \tag{56}$$

holds off that null set, and the exceptional set contributes zero. Integrate using nonnegative additivity and rearrange finite numbers to obtain $\int (f+g)\,d\mu = \int f\,d\mu + \int g\,d\mu$. For a negative scalar, positive and negative parts exchange roles. Consequently all real a, b satisfy

$$\int (af + bg)\,d\mu = a\int f\,d\mu + b\int g\,d\mu. \tag{57}$$

See reference Proposition 2.4.1.

Correction: Undefined pointwise sums

The source writes $\infty + (-\infty) = 0$. This is not an arithmetic identity. For integrable functions one may *choose* zero on the null set where the sum is otherwise undefined, without affecting any integral.

Source page(s): 15–17

22 Proof of MCT: preparatory facts

If $0 \le f \le g$, then $\mathcal{S}_f \subseteq \mathcal{S}_g$, so $\int f\,d\mu \le \int g\,d\mu$. For a fixed nonnegative simple function $s = \sum_{i=1}^{m} a_i \mathbf{1}_{A_i}$ define a *set function*

$$\nu(S) = \int_S s\,d\mu = \sum_{i=1}^{m} a_i \mu(A_i \cap S), \qquad S \in \mathcal{F}. \tag{58}$$

It is a measure: nonnegativity and the empty-set condition are immediate. For disjoint S_j,

$$\nu\left(\bigcup_j S_j\right) = \sum_{i=1}^{m} a_i \sum_j \mu(A_i \cap S_j) \tag{59}$$

$$= \sum_j \sum_{i=1}^{m} a_i \mu(A_i \cap S_j) = \sum_j \nu(S_j). \tag{60}$$

Here all terms are nonnegative. This answers the source's question: S is a measurable *set*; s is the simple *function*.

Source page(s): 18–19

23 Proof of MCT: main step

Monotonicity gives $L := \lim_n \int f_n\,d\mu \le \int f\,d\mu$. Fix $s \in \mathcal{S}_f$ and $0 < \alpha < 1$, and set $S_n = \{\alpha s \le f_n\}$. Then $S_n \uparrow \Omega$: where $s = 0$ the condition holds immediately; where $s > 0$, convergence to $f \ge s > \alpha s$

makes it eventually hold. Continuity from below of the measure ν gives

$$\alpha \int s \, d\mu = \lim_n \int_{S_n} \alpha s \, d\mu \leq \lim_n \int f_n \, d\mu = L. \tag{61}$$

Take the supremum over s and let $\alpha \uparrow 1$ to obtain the reverse inequality.

Original TODO: Why choose alpha strictly below one?

For $f_n = 1 - 1/(n+1)$ and $s = f = 1$, the sets $\{s \leq f_n\}$ are empty for every n. Choosing $\alpha < 1$ gives a strict gap and guarantees eventual inclusion. This is the reason for the extra parameter in reference Theorem 2.3.2.

Source page(s): 20–21

24 Fatou's lemma

For measurable $f_n \geq 0$,

$$\int \liminf_n f_n \, d\mu \leq \liminf_n \int f_n \, d\mu. \tag{62}$$

Set $g_k = \inf_{n \geq k} f_n$. Then $g_k \uparrow \liminf_n f_n$ and, for every $n \geq k$, $g_k \leq f_n$. Therefore

$$\int g_k \, d\mu \leq \inf_{n \geq k} \int f_n \, d\mu. \tag{63}$$

MCT on the left and the definition of liminf on the right prove the assertion (reference Theorem 2.5.1).

Original TODO: Midterm: redo Fatou's proof

The source flags this proof for revision. The essential move is to replace the original sequence by its increasing tail infima, then use MCT. Nonnegativity is the hypothesis that permits MCT.

Correction: Limits cannot be assumed to exist

The handwritten chain uses $\lim_n \int f_n \, d\mu$ before establishing its existence, and ends with "LHS = RHS." Fatou gives an inequality, not generally equality. The tail-infimum comparison above is the valid argument. For strictness, on $\mathbb{R}$ let $f_n = \mathbf{1}_{[n,n+1]}$; then the integral of the liminf is zero and every integral is one.

Source page(s): 22–24

25 Dominated convergence theorem (DCT)

Suppose measurable $f_n \to f$ pointwise and $|f_n| \leq h$ for some nonnegative integrable h. Then f and every f_n are integrable, and

$$\lim_n \int f_n \, d\mu = \int f \, d\mu, \tag{64}$$

$$\lim_n \int |f_n - f| \, d\mu = 0. \tag{65}$$

Proof. We have $|f| \le h$. Apply Fatou to $f_n + h \ge 0$ and subtract the finite number $\int h\,d\mu$:

$$\int f\,d\mu \le \liminf_n \int f_n\,d\mu. \tag{66}$$

Apply the same argument to $-f_n$ to obtain

$$\int f\,d\mu \ge \limsup_n \int f_n\,d\mu. \tag{67}$$

The two bounds prove convergence of the integrals. Apply this result to $|f_n - f| \to 0$, dominated by $2h$, to prove the second assertion (reference Theorem 2.5.2).

Original TODO: Why does convergence in integral absolute difference imply convergence of integrals?

Use

$$\left|\int f_n\,d\mu - \int f\,d\mu\right| \le \int |f_n - f|\,d\mu. \tag{68}$$

The right side tends to zero. This implication does not itself require the full DCT assumptions once integrability and that convergence are given.

Correction: The dominating function must be absolutely integrable

The source sometimes writes only $\int h\,d\mu < \infty$ while placing $|h|$ in the bound. The correct requirement is an integrable nonnegative bound, or equivalently $\int |h|\,d\mu < \infty$. A negative infinite signed integral is not a usable bound.

Source page(s): 25–28

26 Counterexamples and differentiation under the integral

MCT and Fatou in their stated forms require nonnegativity; DCT allows signs.

Original TODO: A counterexample without domination

On $(0,1)$ with Lebesgue measure let $f_n(x) = n\mathbf{1}_{(0,1/n)}(x)$. Then $f_n \to 0$ pointwise and $\int f_n\,d\lambda = 1$. Thus pointwise convergence alone does not allow interchanging a limit and an integral. Domination is a sufficient hypothesis for DCT; it is not a necessary condition for every particular sequence whose integrals happen to converge.

The statistical motivation is a likelihood $\ell_n(\theta) = \sum_{i=1}^n \log p_\theta(X_i)$ for iid data from p_{θ_0}. Under suitable regularity, the score has mean zero:

$$\mathbb{E}_{\theta_0}\left[\frac{\partial}{\partial\theta}\log p_\theta(X)\Big|_{\theta=\theta_0}\right] = \int \frac{\partial p_\theta(x)}{\partial\theta}\,dx\Big|_{\theta=\theta_0} = \frac{\partial}{\partial\theta}\int p_\theta(x)\,dx\Big|_{\theta=\theta_0} = 0. \tag{69}$$

Correction: Regularity for the score argument

The exchange of derivative and integral needs justification, including a common dominating measure, differentiability, and local integrable domination of derivatives. The logarithmic expression also needs positivity almost surely on the relevant support. A zero expected score and a law of large numbers alone do not prove consistency of an arbitrary maximum likelihood estimator; further identification and maximization arguments are needed.

Leibniz rule (reference Proposition 2.5.5). Let $I \subseteq \mathbb{R}$ be open and $F : I \times \Omega \to \mathbb{R}$. Assume each $F(x, \cdot)$ is integrable, each $F(\cdot, \omega)$ is differentiable, and $|F_x(x, \omega)| \leq h(\omega)$ for an integrable nonnegative h. Then

$$\frac{d}{dx} \int F(x, \omega)\, d\mu(\omega) = \int F_x(x, \omega)\, d\mu(\omega). \tag{70}$$

Fix x and $x_n \to x$ with $x_n \neq x$ and eventually in an interval contained in I. The measurable difference quotients

$$g_n(\omega) = \frac{F(x, \omega) - F(x_n, \omega)}{x - x_n} \tag{71}$$

converge to $F_x(x, \omega)$. The mean value theorem gives $|g_n| \leq h$. DCT and linearity identify their integral limit with the derivative of the integral. Since the argument works for every such sequence, the derivative exists as asserted.

Source page(s): 29–30

27 Almost everywhere and almost surely

A property holds *almost everywhere* (a.e.) if it holds outside a measurable null set. In a probability space the same notion is called *almost surely* (a.s.). In particular $f = g$ a.e. means $\{f \neq g\}$ is contained in a measurable null set. If f, g are measurable extended real functions, $\{f \neq g\}$ is measurable, so this is equivalent to $\mu(\{f \neq g\}) = 0$. Without measurability the exceptional set need not itself be measurable. In a complete space every subset of a null set is measurable.

Original TODO: Nonmeasurable functions that agree almost everywhere; what is the difference?

Use the incomplete finite space from Note 02: $\Omega = \{a, b, c\}$ with measurable atoms $\{a, b\}$ (mass zero) and $\{c\}$ (mass one). Let $f = \mathbf{1}_{\{a\}}$ and $g = \mathbf{1}_{\{b\}}$. Both are nonmeasurable, yet $f = g = 0$ outside the measurable null set $\{a, b\}$. Also f agrees a.e. with the measurable zero function, while $\{f \neq 0\} = \{a\}$ is not measurable. This explains why containment in a measurable null set is the general definition. The source references an earlier lecture date here, without specifying another example.

Similarly $f_n \to f$ a.e. means convergence outside one measurable null set, even when measurability of the individual functions has not been assumed (reference Section 2.6).

Source page(s): 30–32

28 Zero integral of a nonnegative function

For measurable $f \geq 0$,

$$\int f \, d\mu = 0 \quad \Longleftrightarrow \quad f = 0 \text{ a.e.} \tag{72}$$

If $f = 0$ a.e., every simple $0 \leq s \leq f$ is supported on a null set and has zero integral. Taking the supremum gives the forward integral assertion. Conversely, if $\mu(\{f > 0\}) > 0$, then

$$\{f > 0\} = \bigcup_{n=1}^{\infty} \{f > 1/n\} \tag{73}$$

implies that some $A_n = \{f > 1/n\}$ has positive measure. Hence

$$\int f \, d\mu \geq \int n^{-1} \mathbf{1}_{A_n} \, d\mu = n^{-1} \mu(A_n) > 0, \tag{74}$$

a contradiction.

Original TODO: Why may an integrable function be infinite somewhere?

Integrability forces its infinite-value set to be null, not empty. For example, $f = \infty \mathbf{1}_{\{0\}}$ on $\mathbb{R}$, interpreted as zero away from zero, is measurable and has integral zero. The convention $0 \cdot \infty = 0$ in integration captures the zero measure of its support. This also explains the source's final "think why" note.

Part IV

STAT 559 – Note 04: Product Measures and Fubini–Tonelli

Source page(s): 1–4

29 Product measurable spaces

The motivation is to turn an integral on a product space into iterated one-dimensional integrals. For measurable spaces $(\Omega_i, \mathcal{F}_i)$, $i = 1, \dots, n$, let

$$\Omega = \prod_{i=1}^{n} \Omega_i, \qquad \mathcal{F} = \bigotimes_{i=1}^{n} \mathcal{F}_i = \sigma\left(\left\{ \prod_{i=1}^{n} A_i : A_i \in \mathcal{F}_i \right\} \right). \tag{75}$$

The generating sets are measurable *rectangles*. When $\Omega_i = \mathbb{R}$ and $\mathcal{F}_i = \mathcal{B}(\mathbb{R})$, this product space is $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$.

Original TODO: Why is the product Borel sigma-algebra the usual one?

For $n = 2$, each Borel rectangle belongs to $\mathcal{B}(\mathbb{R}^2)$: its coordinate factors are inverse images under continuous projections. Conversely every open subset of $\mathbb{R}^2$ is a countable union of open rectangles with rational endpoints, so it belongs to $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$. The two inclusions prove equality. The same countable-base argument works in finite dimensions (reference Exercise 3.1.3).

A desired product measure must satisfy

$$\mu\left(\prod_{i=1}^{n} A_i \right) = \prod_{i=1}^{n} \mu_i(A_i), \qquad 0 \cdot \infty := 0. \tag{76}$$

The plan is to define a premeasure on finite disjoint unions of rectangles, then apply Carathéodory's extension theorem.

Source page(s): 5–6

30 Construction of product measure

Theorem (reference Proposition 3.1.1). If each μ_i is sigma-finite, there exists a unique measure $\mu = \bigotimes_{i=1}^{n} \mu_i$ on $\mathcal{F}$ with the rectangle property. Let $\mathcal{A}$ be the finite disjoint unions of measurable rectangles. Define

$$\mu_0\left(\bigsqcup_{j=1}^{m} \prod_{i=1}^{n} A_{j,i} \right) = \sum_{j=1}^{m} \prod_{i=1}^{n} \mu_i(A_{j,i}). \tag{77}$$

Supplement: Algebra and well-definedness homework

Intersecting rectangles gives a rectangle. A difference of rectangles splits into finitely many disjoint rectangles by selecting the first coordinate that lies outside the second rectangle. This yields complements and finite unions in $\mathcal{A}$. A common finite refinement of any two representations, together with finite additivity in each coordinate, shows that the displayed value is independent of the representation.

Source page(s): 7–10

31 Countable additivity by dimension reduction

It suffices to prove the identity when a rectangle is partitioned into countably many rectangles:

$$\prod_{j=1}^{n} A_j = \bigsqcup_{i=1}^{\infty} \prod_{j=1}^{n} A_{i,j}. \tag{78}$$

Induct on n, the case $n = 1$ being ordinary countable additivity. Fix $x \in \prod_{j<n} A_j$ and let

$$I_x = \left\{ i : x \in \prod_{j<n} A_{i,j} \right\}. \tag{79}$$

Taking the section at x gives $A_n = \bigsqcup_{i \in I_x} A_{i,n}$. Consequently

$$\mathbf{1}_{\prod_{j<n} A_j}(x)\mu_n(A_n) = \sum_{i=1}^{\infty} \mathbf{1}_{\prod_{j<n} A_{i,j}}(x)\mu_n(A_{i,n}). \tag{80}$$

If x lies outside the left rectangle's first $n-1$ factors, every active right rectangle has empty last factor; thus the identity holds there too, with the zero-times-infinity convention. Integrate with respect to the already constructed $\mu' = \bigotimes_{j<n} \mu_j$ and apply MCT to the nonnegative series. The rectangle property in dimension $n-1$ yields

$$\prod_{j=1}^{n} \mu_j(A_j) = \sum_{i=1}^{\infty} \prod_{j=1}^{n} \mu_j(A_{i,j}). \tag{81}$$

This proves the required premeasure property. Carathéodory gives existence. Choose increasing finite-measure exhaustions in each coordinate; their product rectangles exhaust Ω and have finite mass. The uniqueness theorem then applies to the pi-system of rectangles.

Supplement: The source's finite-measure reduction

The notes say to assume the measures finite without elaboration. One may localize to the finite-measure rectangles from the exhaustions and pass to an increasing limit. The section proof above also works directly with nonnegative extended quantities, provided products with zero are interpreted consistently.

Source page(s): 11–12

32 Fubini–Tonelli theorem

Let $(\Omega_i, \mathcal{F}_i, \mu_i)$, $i = 1, 2$, be sigma-finite and let $\mu = \mu_1 \otimes \mu_2$. Let $f : \Omega_1 \times \Omega_2 \to \overline{\mathbb{R}}$ be product-measurable. If $f \geq 0$, the section integrals are measurable extended nonnegative functions and

$$\int f \, d\mu = \int_{\Omega_1} \left(\int_{\Omega_2} f(x, y) \, d\mu_2(y) \right) d\mu_1(x) \tag{82}$$

$$= \int_{\Omega_2} \left(\int_{\Omega_1} f(x, y) \, d\mu_1(x) \right) d\mu_2(y). \tag{83}$$

This is commonly called *Tonelli's theorem*. If $\int |f| \, d\mu < \infty$, the same equality holds with finite integrals; the sections are integrable for almost every outer coordinate. This is *Fubini's theorem*. On the exceptional outer-coordinate sets where a signed section integral is undefined, set that integral to zero. Moreover, either finite iterated integral of $|f|$ proves that f is integrable, by the nonnegative case (reference Theorem 3.2.2).

Original TODO: What is meant by product-measurable?

It means $f^{-1}(C) \in \mathcal{F}_1 \otimes \mathcal{F}_2$ for every Borel $C \subseteq \overline{\mathbb{R}}$. Measurability of each separate section alone is a different hypothesis and does not in general imply this. The reference combines the nonnegative and integrable cases under the name Fubini's theorem; the source's attribution of the names is retained here with the common distinction explained.

Source page(s): 13

Original TODO: Check section measurability

For fixed x, the sets E whose sections $E_x = \{y : (x, y) \in E\}$ lie in $\mathcal{F}_2$ form a sigma-algebra containing the rectangles. Hence every product-measurable set has measurable sections. Applying this to inverse images of Borel sets shows $y \mapsto f(x, y)$ is measurable (reference Lemma 3.2.1). The other coordinate is symmetric.

Source page(s): 14–16

33 Proof: rectangles, simple functions, and limits

For $f = \mathbf{1}_{A_1 \times A_2}$, direct computation gives

$$\int_{\Omega_1} \int_{\Omega_2} \mathbf{1}_{A_1 \times A_2}(x, y) \, d\mu_2(y) \, d\mu_1(x) = \mu_1(A_1)\mu_2(A_2) = \mu(A_1 \times A_2). \tag{84}$$

After extending the indicator identity to all measurable sets (next section), finite nonnegative linear combinations prove it for simple functions. If $s_n \uparrow f \geq 0$, MCT first in the inner integral, then in the outer integral, and also on the product space proves the identity for f. The section-integral functions are measurable as increasing limits of the corresponding simple-function integrals.

Source page(s): 17–20

34 Proof: general indicators via the pi–lambda theorem

For finite μ_1, μ_2, let $\mathcal{D}$ consist of product-measurable sets E for which $x \mapsto \mu_2(E_x)$ is measurable and

$$\int_{\Omega_1} \mu_2(E_x)\, d\mu_1(x) = \mu(E). \tag{85}$$

Rectangles belong to $\mathcal{D}$. If $E_i \in \mathcal{D}$ are disjoint, their sections are disjoint, so

$$\mu_2\left(\left(\bigcup_i E_i\right)_x\right) = \sum_i \mu_2((E_i)_x), \tag{86}$$

$$\int \sum_i \mu_2((E_i)_x)\, d\mu_1(x) = \sum_i \mu(E_i) = \mu\left(\bigcup_i E_i\right). \tag{87}$$

Measurability follows from nonnegative measurable summation and the integral equality from MCT. For complements,

$$\mu_2((E^c)_x) = \mu_2(\Omega_2) - \mu_2(E_x). \tag{88}$$

All terms are finite, and integration gives $\mu(\Omega) - \mu(E) = \mu(E^c)$. Thus $\mathcal{D}$ is a lambda-system containing the generating pi-system of rectangles. Dynkin's theorem proves the indicator identity.

Original TODO: Review the pi–lambda proof technique

The defining class must include both the measurability of the section-mass map and its integral identity. Establish the result on the easy generating pi-system, prove the desired class is a lambda-system, then invoke Dynkin. Merely declaring the class to be a sigma-algebra would skip the substantive step.

Supplement: Passing from finite to sigma-finite measures

Choose $X_n \uparrow \Omega_1$ and $Y_n \uparrow \Omega_2$ with finite coordinate measures. The restricted measures $\mu_{1,n}(A) = \mu_1(A \cap X_n)$ and $\mu_{2,n}(B) = \mu_2(B \cap Y_n)$ have product $E \mapsto \mu(E \cap (X_n \times Y_n))$, by equality on rectangles and uniqueness. Apply the finite result and then MCT to

$$x \longmapsto \mathbf{1}_{X_n}(x)\mu_2(E_x \cap Y_n) \uparrow \mu_2(E_x). \tag{89}$$

This supplies the localization step omitted in the handwritten proof (reference proof of Theorem 3.2.2).

Original TODO: Review the integrable case

Apply the nonnegative theorem to $|f|$ to see that $\int |f(x,y)|\, d\mu_2(y) < \infty$ for almost every x. Apply it to f^+ and f^- and subtract their finite total integrals. Their section integrals are finite outside a common null set; hence subtraction there is legitimate. Repeat in the other order. This justifies the source's abbreviated signed-integral step without requiring every section to be integrable.

Source page(s): 21–22

35 Midterm preparation recorded in the source

The handwritten announcement describes four questions: a first true/false question and three further questions with one to four subparts. Coverage is Chapters 1 through 3.2. Listed preparation topics are:

1. Decide whether a set function is a measure and whether it is sigma-finite.
2. Determine the sigma-algebra generated by a specified class.
3. Check measurability and compute Lebesgue integrals on simple spaces.
4. Decide whether a limit can be exchanged with an integral using MCT, Fatou, or DCT.
5. Decide whether an iterated integral equals the product-space integral for a concrete function.
6. Distinguish Riemann integration from improper integration.

The source describes the exam as easier than the homework. This is a transcription of the note's announcement, not a verification of a current exam policy.

Part V

STAT 559 – Note 05: Inequalities and the Foundations of Lp Spaces

Source page(s): 1

36 Markov's inequality

For a nonnegative measurable f on any measure space and $t > 0$,

$$\mu(\{f \geq t\}) \leq \frac{1}{t} \int f \, d\mu. \tag{90}$$

For a nonnegative random variable this reads $\mathbb{P}(X \geq t) \leq \mathbb{E}X/t$.

Original TODO: Prove Markov's inequality

The pointwise inequality $t\mathbf{1}_{\{f \geq t\}} \leq f$ gives the result immediately after integration and division by $t > 0$ (reference Theorem 4.1.1).

Source page(s): 1–2

37 Jensen's and Young's inequalities

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let f be integrable with values in an interval I, and let $\phi : I \to \mathbb{R}$ be convex. When $\phi \circ f$ is integrable,

$$\phi\left(\int f \, d\mathbb{P}\right) \leq \int \phi(f) \, d\mathbb{P}, \tag{91}$$

or $\phi(\mathbb{E}X) \leq \mathbb{E}[\phi(X)]$.

Correction: Integrability in the stated form of Jensen

The handwritten theorem does not state the condition on $\phi \circ f$. The finite-integral version in reference Theorem 4.2.6 assumes it is integrable. One can also formulate an extended-value version with a well-defined right side, but the finite version suffices here.

Original TODO: Prove Jensen's inequality

Put $m = \int f \, d\mathbb{P}$, which belongs to I. If m is an included endpoint of I, nonnegativity of $f - m$ or $m - f$ and zero integral show $f = m$ a.e. Otherwise a supporting affine function exists: $\phi(x) \geq ax + b$ on I and $\phi(m) = am + b$. Integrating gives $\int \phi(f) \, d\mathbb{P} \geq am + b = \phi(m)$. See reference Section 4.2.

For conjugate exponents $p, q \in (1, \infty)$ with $1/p + 1/q = 1$, Young's inequality is

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q} \qquad (x, y \geq 0). \tag{92}$$

Original TODO: Prove Young's inequality

For positive x, y, concavity of the logarithm gives

$$\log\left(\frac{x^p}{p}+\frac{y^q}{q}\right) \geq \frac{1}{p}\log(x^p)+\frac{1}{q}\log(y^q)=\log(xy). \tag{93}$$

Exponentiate. If either variable is zero the inequality is immediate (reference Exercise 4.2.8). The source's phrase "natural couple" means conjugate exponents.

Source page(s): 3–5

38 Lp norms and the completeness goal

For measurable $f:\Omega\to\overline{\mathbb{R}}$ and $1\leq p<\infty$, define

$$\|f\|_p=\left(\int |f|^p\,d\mu\right)^{1/p}. \tag{94}$$

For $p=\infty$ define the essential supremum

$$\|f\|_\infty=\inf\{K\in[0,\infty]:|f|\leq K \text{ a.e.}\}. \tag{95}$$

The finite-norm measurable functions, identified up to a.e. equality, form $L^p(\Omega,\mathcal{F},\mu)$. The goal is to prove this is a normed vector space and is complete, hence a Banach space. A sequence is Cauchy if for every $\varepsilon>0$ there exists N such that $\|f_n-f_m\|_p<\varepsilon$ whenever $n,m\geq N$.

Correction: A norm is defined on equivalence classes

On literal functions, $\|f\|_p=0$ means $f=0$ a.e., not necessarily everywhere. Thus the integral formula is only a seminorm before quotienting by a.e. equality. The quotient convention is necessary for the source's norm and Banach-space claims (reference Section 4.4).

Source page(s): 6–7

A real vector space has associative and commutative addition, a zero element, additive inverses, and scalar multiplication satisfying

$$a(bu)=(ab)u, \qquad 1u=u, \tag{96}$$

$$a(u+v)=au+av, \qquad (a+b)u=au+bu. \tag{97}$$

A norm is a finite nonnegative function on a vector space satisfying

$$\|u+v\|\leq\|u\|+\|v\|, \tag{98}$$

$$\|au\|=|a|\|u\|, \tag{99}$$

$$\|u\|=0\iff u=0. \tag{100}$$

Correction: Missing vector-space axiom

The source lists seven rules but omits $1u = u$. It is included above. Closure of addition and scalar multiplication is part of specifying the operations on the vector space.

The triangle inequality for L^p is Minkowski's inequality; its proof uses Hölder's inequality.

Source page(s): 8–9

39 Hölder's inequality

For conjugate exponents $1 \le p, q \le \infty$, with $1/\infty = 0$,

$$\|fg\|_1 \le \|f\|_p \|g\|_q. \tag{101}$$

At $p = q = 2$ this is Cauchy–Schwarz. For $p = 1, q = \infty$, use $|g| \le \|g\|_\infty$ a.e. to obtain

$$\int |fg| \, d\mu \le \|g\|_\infty \int |f| \, d\mu. \tag{102}$$

The opposite endpoint follows by symmetry.

Original TODO: Verify the essential-supremum bound

If $M = \|g\|_\infty < \infty$, for each n there is an a.e. bound less than $M + 1/n$. Outside the union of their exceptional null sets, $|g| \le M + 1/n$ for all n, hence $|g| \le M$. This justifies the endpoint argument.

For $1 < p, q < \infty$, first dispose of zero norms (the corresponding function is zero a.e.) and infinite right sides. For positive finite norms set $u = f/\|f\|_p$ and $v = g/\|g\|_q$. Young's inequality and integration give

$$\int |uv| \, d\mu \le \frac{1}{p} \int |u|^p \, d\mu + \frac{1}{q} \int |v|^q \, d\mu = \frac{1}{p} + \frac{1}{q} = 1. \tag{103}$$

Rescaling proves the claim (reference Theorem 4.4.1).

Correction: The normalization of the second function

The source's denominator for v resembles $\|g\|_p$. The required denominator is $\|g\|_q$, so $\int |v|^q \, d\mu = 1$. Division also requires checking that the norm is positive and finite first.

Source page(s): 10–11

40 Minkowski's inequality

For $1 \le p \le \infty$,

$$\|f + g\|_p \le \|f\|_p + \|g\|_p. \tag{104}$$

At $p = 1$ integrate $|f + g| \le |f| + |g|$.

Original TODO: Verify the infinity case using null sets

Outside the union of the null sets where the two essential bounds fail,

$$|f+g| \le |f| + |g| \le \|f\|_\infty + \|g\|_\infty. \tag{105}$$

The definition of essential supremum gives the desired result. Ordinary pointwise suprema, suggested in the handwritten margin, do not ignore exceptional null sets and need not equal these norms.

For $1 < p < \infty$, it suffices to consider finite right side. First, convexity gives

$$|f+g|^p \le 2^{p-1}(|f|^p + |g|^p), \tag{106}$$

so $\|f+g\|_p$ is finite. Hölder gives

$$\|f+g\|_p^p \le \int |f||f+g|^{p-1}\,d\mu + \int |g||f+g|^{p-1}\,d\mu \tag{107}$$

$$\le (\|f\|_p + \|g\|_p)\,\big\||f+g|^{p-1}\big\|_q. \tag{108}$$

Original TODO: Verify the exponent simplification

Since $q = p/(p-1)$,

$$\big\||f+g|^{p-1}\big\|_q = \left(\int |f+g|^{(p-1)q}\,d\mu\right)^{1/q} = \|f+g\|_p^{p/q} = \|f+g\|_p^{p-1}. \tag{109}$$

If $\|f+g\|_p = 0$ there is nothing to prove. Otherwise divide by its finite positive $(p-1)$st power. The preliminary finiteness argument avoids an invalid cancellation of infinities (reference Theorem 4.4.3).

Source page(s): 12

41 First Borel–Cantelli lemma

If $\sum_n \mu(A_n) < \infty$, then

$$\mu(\limsup_n A_n) = 0, \qquad \limsup_n A_n = \bigcap_{m=1}^{\infty} \bigcup_{n \ge m} A_n. \tag{110}$$

Thus almost every point belongs to only finitely many A_n.

Supplement: Proof used in the completeness argument

For every m, the union bound gives $\mu(\limsup_n A_n) \le \sum_{n \ge m} \mu(A_n)$. Tails of a convergent nonnegative series tend to zero, proving the lemma. No independence is required (reference Theorem 4.3.1).

Source page(s): 13–15

42 An almost-everywhere convergent subsequence

Lemma. Every Cauchy sequence in L^p, $1 \le p \le \infty$, has a subsequence converging a.e. to a measurable function $f \in L^p$. For $p < \infty$, choose increasing indices n_k with $\|f_{n_k} - f_{n_{k+1}}\|_p \le 2^{-k}$. Define

$$A_k = \{|f_{n_k} - f_{n_{k+1}}| \ge 2^{-k/2}\}. \tag{111}$$

Markov applied to the pth power gives

$$\mu(A_k) \le 2^{kp/2} \|f_{n_k} - f_{n_{k+1}}\|_p^p \le 2^{-kp/2}. \tag{112}$$

The series is summable, so $N = \limsup_k A_k$ is null. Outside N the successive differences are eventually at most $2^{-k/2}$.

Correction: The good set is the complement of the limsup

Page 15 calls the infinitely-often set B, proves $\mu(B) = 0$, then writes the eventual difference bound for points in B. That bound holds on B^c. We use N for the null bad set to avoid the conflicting conventions on pages 14–15.

Source page(s): 16

Original TODO: Fill in the remaining argument; recording marker

The only writing on this page asks to "fill in" material and appears to include a recording timestamp around 35:40. The exact digits are uncertain and no recording is among the permitted sources. The mathematical continuation can nevertheless be supplied from reference Lemma 4.4.5: for $l > k$ sufficiently large and $\omega \notin N$,

$$|f_{n_l}(\omega) - f_{n_k}(\omega)| \le \sum_{j=k}^{l-1} 2^{-j/2} \longrightarrow 0 \quad (k \to \infty). \tag{113}$$

Thus the subsequence is pointwise Cauchy outside a measurable null set and has a finite limit there; set the limit equal to zero on that set. Choose finite measurable representatives of the original L^p classes. The limit so defined is measurable, and Fatou gives

$$\int |f|^p \, d\mu \le \liminf_k \|f_{n_k}\|_p^p < \infty, \tag{114}$$

since Cauchy sequences are norm-bounded. For $p = \infty$ choose the same subsequence in essential-supremum norm, remove the countable union of exceptional sets, and use $|f_{n_{k+1}} - f_{n_k}| \le 2^{-k}$ on the remaining set. The telescoping series gives convergence and $|f| \le |f_{n_1}| + \sum_k 2^{-k}$ a.e., so $f \in L^\infty$. The completion of the Banach-space argument is supplied below.

Supplement: Completing the completeness theorem

The source stops at the subsequence lemma. To finish its stated goal, let the whole sequence be Cauchy and take the a.e. limit f of the subsequence just constructed. For $p < \infty$, if $n \geq N_\varepsilon$, Fatou applied as $k \to \infty$ gives

$$\|f_n - f\|_p^p \leq \liminf_k \|f_n - f_{n_k}\|_p^p \leq \varepsilon^p. \tag{115}$$

Thus the full sequence converges to f in norm. For $p = \infty$, the eventual bounds $|f_n - f_{n_k}| \leq \varepsilon$ hold outside a countable union of null sets; take the pointwise subsequence limit to obtain $\|f_n - f\|_\infty \leq \varepsilon$. This proves completeness for all the stated exponents (reference Theorem 4.4.6).

Part VI

STAT 559 – Note 6: Random Variables, Laws, and Independence

Source page(s): 1–2

43 Random variables and generated information

An Ω'-valued random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ is a measurable map into a measurable space $(\Omega', \mathcal{F}')$. We primarily consider real-valued random variables, with target $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. The event notation is

$$\{X \in A\} = \{\omega : X(\omega) \in A\}, \tag{116}$$

$$\{X \in A, Y \in B\} = \{\omega : X(\omega) \in A,\ Y(\omega) \in B\}. \tag{117}$$

For measurable $g : \mathbb{R} \to \mathbb{R}$, $g(X)$ means the composition $g \circ X$, which is again measurable. The sigma-algebra generated by X is

$$\sigma(X) = \{X^{-1}(A) : A \in \mathcal{B}(\mathbb{R})\}. \tag{118}$$

It is the smallest sigma-algebra on Ω making X measurable. For any index set I, possibly uncountable,

$$\sigma(X_i : i \in I) = \sigma\left(\bigcup_{i \in I} \sigma(X_i)\right). \tag{119}$$

For a fair coin, take $\Omega = \{H, T\}$, $\mathcal{F} = 2^{\Omega}$, $\mathbb{P}(\{H\}) = \mathbb{P}(\{T\}) = 1/2$, and $X(H) = 1, X(T) = 0$. Then $\mathbb{P}(X = 1) = 1/2$.

Source page(s): 3

44 The law of a random variable

The *law* of X is the induced probability measure on the target:

$$\mu_X(A) = \mathbb{P}(X \in A) = \mathbb{P}(X^{-1}(A)), \qquad A \in \mathcal{B}(\mathbb{R}). \tag{120}$$

It is also denoted P_X; the notation $X \sim \mu_X$ specifies this distribution. For example, take the probability space

$$(\Omega, \mathcal{F}, \mathbb{P}) = ([0,1], \mathcal{B}([0,1]), \lambda), \qquad X(\omega) = \omega^2. \tag{121}$$

The sample-space measure is uniform, while the law of X is its pushforward under squaring.

Correction: Where the randomness is represented

The source says that $\mathbb{P}$ has "nothing to do" with the randomness of X. The function X itself is deterministic; its probabilistic behavior is determined jointly by X and $\mathbb{P}$ through μ_X. In this example its cdf on $[0,1]$ is $\sqrt{t}$, not the uniform cdf. No additional randomness axiom is needed (reference Sections 5.1 and 5.3).

Source page(s): 4–5

45 Cumulative distribution functions

Define

$$F_X(t) = \mathbb{P}(X \leq t) = \mu_X((-\infty, t]). \tag{122}$$

A function $F : \mathbb{R} \to [0, 1]$ is a cdf exactly when it is nondecreasing, right-continuous, and satisfies

$$\lim_{t \to -\infty} F(t) = 0, \qquad \lim_{t \to \infty} F(t) = 1. \tag{123}$$

Correction: The missing monotonicity condition

The source leaves condition (a) blank. Nondecreasingness is required, in addition to right-continuity and the two limits. The source calls this a Skorokhod theorem; here it is the inverse-cdf construction of reference Proposition 5.2.2, distinct from the later Skorokhod representation theorem for weak convergence.

Given such F, take $\Omega = (0, 1)$ with Lebesgue probability measure and define

$$X(u) = F^{-1}(u) := \inf\{t \in \mathbb{R} : F(t) \geq u\}. \tag{124}$$

The endpoint limits make this finite for $0 < u < 1$. Monotonicity and right-continuity yield $X(u) \leq t$ if and only if $u \leq F(t)$. Thus X is measurable and $\mathbb{P}(X \leq t) = F(t)$. Conversely monotonicity of probabilities and continuity of a measure along increasing and decreasing sequences establish the cdf properties.

Correction: Avoiding inverse-cdf endpoint ambiguities

The source uses $[0, 1]$. The infimum formula can produce infinite values at zero or one. Using $(0, 1)$, as in the reference, avoids this; alternatively assign arbitrary finite values at these two null endpoints.

Source page(s): 5–6

46 The cdf determines the law

Two real random variables have the same cdf if and only if they have the same law. One direction follows by evaluating the laws on $(-\infty, t]$. For the other define

$$\mathcal{D} = \{(-\infty, t] : t \in \mathbb{R}\}, \qquad \mathcal{G} = \{A \in \mathcal{B}(\mathbb{R}) : \mu_X(A) = \mu_Y(A)\}. \tag{125}$$

Then $\mathcal{D}$ is a pi-system generating $\mathcal{B}(\mathbb{R})$ and equal cdfs imply $\mathcal{D} \subseteq \mathcal{G}$.

Original TODO: Check the lambda-system step

The whole space lies in $\mathcal{G}$ since both laws have mass one. If $A \in \mathcal{G}$, then $\mu_X(A^c) = 1 - \mu_X(A) = 1 - \mu_Y(A) = \mu_Y(A^c)$. Countable additivity gives closure under disjoint countable unions. Hence $\mathcal{G}$ is a lambda-system, and Dynkin's theorem proves equality on all Borel sets (reference Proposition 5.3.1).

Source page(s): 7–8

47 Densities and absolute continuity

For two measures, $\nu \ll \mu$ means $\mu(A) = 0 \Longrightarrow \nu(A) = 0$ for every measurable A. A probability law admitting a density with respect to Lebesgue measure satisfies

$$\mu_X(A) = \int_A f \, d\lambda, \qquad f \geq 0, \quad \int_{\mathbb{R}} f \, d\lambda = 1. \tag{126}$$

The density is denoted $d\mu_X/d\lambda$, a Radon–Nikodym derivative. Conversely such a nonnegative measurable normalized f defines a probability law by this formula. Reference Section 5.4 develops the density definition directly.

Supplement: The source's optional decomposition remark

For a finite measure ν on $\mathbb{R}$, the Lebesgue decomposition relative to λ has the form $\nu = \nu_{\text{ac}} + \nu_{\text{s}}$ with $\nu_{\text{ac}} \ll \lambda$ and $\nu_{\text{s}} \perp \lambda$. Singularity means the singular part is concentrated on a Lebesgue-null measurable set. This is a side remark in the handwriting, not proved in the provided reference's density section. The unrestricted phrase "every measure" in the source requires general measure-theoretic hypotheses; the finite-law case is the one relevant here.

To verify a candidate density it suffices to check its integral formula on closed intervals $[a, b]$ whose endpoints are continuity points of F_X. With f a nonnegative measurable candidate, increasing such intervals to $\mathbb{R}$ first shows normalization. Continuity-point approximation extends the equality to arbitrary intervals, then the pi–lambda theorem gives all Borel sets (reference Proposition 5.4.1).

Source page(s): 9

Supplement: Density homework

If f, g are densities for the same law, $\int_A (f - g) \, d\lambda = 0$ for every Borel A. Applying this to $\{f > g\}$ and $\{f < g\}$ proves $f = g$ a.e. Also the cdf associated to f is

$$F(t) = \int_{-\infty}^{t} f(y) \, d\lambda(y). \tag{127}$$

The integral is unchanged by including the endpoint because singletons are Lebesgue-null. See reference Exercises 5.4.2–5.4.3. The source assigns Sections 5.5–5.6 (standard continuous and discrete distributions) as self-reading.

Source page(s): 10–11

48 Expectation and changing to the law

Expectation is a Lebesgue integral:

$$\mathbb{E}X = \int_{\Omega} X(\omega) \, d\mathbb{P}(\omega), \tag{128}$$

provided the positive and negative parts do not both have infinite integral. The Cauchy distribution is an example with undefined mean. For $X \sim N(m, \sigma^2)$ the expectation is m. The *law of the*

unconscious statistician states that for Borel g, whenever the integral is defined,

$$\mathbb{E}[g(X)] = \int_\Omega g(X(\omega))\, d\mathbb{P}(\omega) = \int_{\mathbb{R}} g(x)\, d\mu_X(x). \tag{129}$$

If μ_X has density f_X, this equals $\int_{\mathbb{R}} g(x) f_X(x)\, d\lambda(x)$.

Supplement: Why this change of space is valid

For indicator $g = \mathbf{1}_A$ this is the definition of μ_X. Extend by finite sums to nonnegative simple functions, by MCT to nonnegative measurable functions, then by positive and negative parts to signed functions with defined integral (reference Section 6.1).

Supplement: Tail-integral homework

For nonnegative X, Tonelli applied to $\mathbf{1}_{\{0<t<X(\omega)\}}$ gives

$$\mathbb{E}X = \int_0^\infty \mathbb{P}(X > t)\, dt. \tag{130}$$

Indeed $X(\omega) = \int_0^\infty \mathbf{1}_{\{t<X(\omega)\}}\, dt$. The source writes $\mathbb{P}(X \geq t)$; that has the same integral because the distribution has only countably many atoms. Both formulas allow infinity (reference Section 6.1).

The source emphasizes studying the *characteristic function* for the final. Its definition is $\varphi_X(t) = \mathbb{E}(e^{itX})$, and it always exists because the integrand has absolute value one (reference Section 6.4).

Source page(s): 12–13

49 Independence of sigma-algebras, events, and variables

Sub-sigma-algebras $\mathcal{F}_1, \ldots, \mathcal{F}_n \subseteq \mathcal{F}$ are independent if for every $A_i \in \mathcal{F}_i$,

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n \mathbb{P}(A_i). \tag{131}$$

Events $A_1, \ldots, A_n$ are independent when their generated sigma-algebras are independent. Explicitly,

$$\sigma(A_i) = \{\varnothing, \Omega, A_i, A_i^c\}, \tag{132}$$

with duplicate sets removed. An arbitrary family of sigma-algebras is independent if every finite subfamily is independent. Random variables $\{X_i\}_{i\in I}$ are independent if $\{\sigma(X_i)\}_{i\in I}$ are independent.

Original TODO: Why is sigma(X) interpreted as information?

Its events are precisely the events of the form $\{X \in B\}$ that can be determined by observing X. For the coin example these distinguish heads from tails; a constant random variable generates only $\{\varnothing, \Omega\}$. Independence of these sigma-algebras formalizes that any finitely many observable events across the separate variables have probabilities that factor. See reference Sections 5.1 and 7.1.

Source page(s): 14–15

50 Connecting the elementary definition and Borel–Cantelli

For Bernoulli variables, independence can be checked by factoring all four joint point masses:

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y), \qquad x, y \in \{0, 1\}. \tag{133}$$

Summing over the relevant values gives the event-based definition. In general the equivalent statement is that the joint law is the product of the marginal laws; the symbols $P(X, Y) = P(X)P(Y)$ in the source abbreviate this measure identity. The first Borel–Cantelli lemma says

$$\sum_n \mathbb{P}(A_n) < \infty \quad \Longrightarrow \quad \mathbb{P}(A_n \text{ infinitely often}) = 0. \tag{134}$$

The second says that for independent events,

$$\sum_n \mathbb{P}(A_n) = \infty \quad \Longrightarrow \quad \mathbb{P}(A_n \text{ infinitely often}) = 1. \tag{135}$$

Supplement: Proof of the second lemma

For fixed $m \le n$, independence and $1 - u \le e^{-u}$ imply

$$\mathbb{P}\left(\bigcap_{k=m}^{n} A_k^c\right) = \prod_{k=m}^{n} (1 - \mathbb{P}(A_k)) \le \exp\left(-\sum_{k=m}^{n} \mathbb{P}(A_k)\right) \longrightarrow 0. \tag{136}$$

Hence the probability of no occurrences after any fixed m is zero. The event of finitely many occurrences is the countable union of these null events, so it is null (reference Theorem 7.3.1).

Correction: Independence is needed for the zero–one conclusion here

Together the two lemmas show that for an independent sequence, the infinitely-often event has probability zero or one according to convergence or divergence of the probability sum. Without independence it may have any probability: taking $A_n = A$ for every n gives probability $\mathbb{P}(A)$. Page 14's sum must refer to the same probability measure as its conclusion, rather than an unrelated μ.

Source page(s): 16

51 Optional topics recorded in the source

The final page lists the Kolmogorov zero–one law and the construction of an infinite product $P_1 \otimes P_2 \otimes \cdots$ as optional topics. The supplied reference treats these in Sections 7.4 and 3.3 respectively. The source contains no proof on this page.

Part VII

STAT 559 – Note 7: Modes of Convergence and Uniform Integrability

Source page(s): 1–3

52 Four modes of convergence

Unless otherwise specified, X_n and X are random variables on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$. *Almost sure convergence*, written $X_n \to X$ a.s., means

$$\mathbb{P}(\{\omega : \lim_n X_n(\omega) = X(\omega)\}) = 1. \tag{137}$$

Convergence in probability, written $X_n \xrightarrow{\mathbb{P}} X$, means

$$\mathbb{P}(|X_n - X| > \varepsilon) \longrightarrow 0 \quad \text{for every } \varepsilon > 0. \tag{138}$$

Convergence in L^p, for $1 \le p \le \infty$, means $X_n, X \in L^p$ and

$$\|X_n - X\|_p \longrightarrow 0. \tag{139}$$

For finite p this is equivalently $\mathbb{E}|X_n - X|^p \to 0$; $p = 1$ is convergence in mean. *Convergence in distribution*, also called convergence in law or weak convergence and written $X_n \Rightarrow X$ or $X_n \xrightarrow{d} X$, means

$$F_{X_n}(t) \longrightarrow F_X(t) \quad \text{at every continuity point } t \text{ of } F_X. \tag{140}$$

For this last notion the variables may live on different probability spaces; only their laws are compared. Convergence is not required at discontinuities of the limiting cdf (reference Section 8.1).

Correction: Which measure tests almost sure convergence?

The first page writes a null-set expression using μ_X. The convergence event is a subset of the sample space Ω, so it is measured by $\mathbb{P}$, not by the law μ_X on $\mathbb{R}$.

Source page(s): 4–5

53 Almost sure convergence implies convergence in probability

Fix $\varepsilon > 0$ and put $S_n = \bigcap_{k \ge n} \{|X_k - X| \le \varepsilon\}$. Almost sure convergence implies $\mathbb{P}(\bigcup_n S_n) = 1$. Since S_n increases and $S_n \subseteq \{|X_n - X| \le \varepsilon\}$,

$$1 = \lim_n \mathbb{P}(S_n) \le \liminf_n \mathbb{P}(|X_n - X| \le \varepsilon) \le 1. \tag{141}$$

The complement probabilities tend to zero.

Original TODO: Translate the limit statement into events

"Eventually within ε" means "there exists n such that for every $k \geq n$ the bound holds." The existential quantifier becomes a union and the universal quantifier becomes an intersection. To express convergence itself, use a countable family of tolerances such as $1/m$ and intersect over m. Interchanging an uncountable family of probability-one events would require care; the countable tolerances avoid that issue (reference Proposition 8.2.1).

Source page(s): 5–7

54 Convergence in probability has an a.s. convergent subsequence

If $X_n \xrightarrow{\mathbb{P}} X$, choose increasing indices n_k such that

$$\mathbb{P}(|X_{n_k} - X| > 2^{-k-1}) \leq 2^{-k}. \tag{142}$$

The thresholds decrease, so both n_k and n_{k+1} can be chosen late enough for the bound at tolerance 2^{-k-1}. The triangle inequality gives

$$\mathbb{P}(|X_{n_k} - X_{n_{k+1}}| > 2^{-k}) \leq 2 \cdot 2^{-k}. \tag{143}$$

The constant factor does not affect summability. Borel–Cantelli and telescoping, as in the L^p completeness proof, show that X_{n_k} is pointwise Cauchy outside a null set. It converges there to a measurable Y; set $Y = 0$ on the exceptional set. It remains to identify $Y = X$ a.s.

Original TODO: Final-exam candidate: uniqueness of the probability limit

Almost sure convergence to Y implies convergence in probability to Y. For any $\varepsilon > 0$,

$$\mathbb{P}(|X - Y| > \varepsilon) \leq \mathbb{P}(|X - X_{n_k}| > \varepsilon/2) + \mathbb{P}(|X_{n_k} - Y| > \varepsilon/2) \longrightarrow 0. \tag{144}$$

Taking $\varepsilon = 1/m$ for all positive integers m proves $X = Y$ a.s. Thus the subsequence converges a.s. to X (reference Proposition 8.2.3).

Supplement: The source's Riesz–Fischer connection

The source reuses the Cauchy-subsequence technique from the Riesz–Fischer proof. The result here is the subsequence principle for convergence in probability; Riesz–Fischer itself states completeness of L^p. A handwritten reminder "fill in: 12:12 pm" appears beside the proof, but no corresponding recording is supplied. The complete argument above covers the mathematical gap. One can also prove the claim directly by Borel–Cantelli on $\{|X_{n_k} - X| > 2^{-k}\}$.

Source page(s): 8–9

55 Convergence in probability implies convergence in law

For $\varepsilon > 0$ the event inclusions give both bounds

$$F_{X_n}(t) \leq F_X(t + \varepsilon) + \mathbb{P}(|X_n - X| > \varepsilon), \tag{145}$$

$$F_{X_n}(t) \geq F_X(t - \varepsilon) - \mathbb{P}(|X_n - X| > \varepsilon). \tag{146}$$

The first follows because $X_n \le t$ and $|X_n - X| \le \varepsilon$ imply $X \le t + \varepsilon$. The second follows because $X \le t - \varepsilon$ and the same closeness imply $X_n \le t$. Taking limsup and liminf in n yields

$$F_X(t-\varepsilon) \le \liminf_n F_{X_n}(t) \le \limsup_n F_{X_n}(t) \le F_X(t+\varepsilon). \tag{147}$$

At a continuity point t let $\varepsilon \downarrow 0$; the outer terms both tend to $F_X(t)$, proving the result (reference Proposition 8.2.4).

Correction: The lower-bound step

The source says to obtain the other bound by choosing a negative epsilon. The error tolerance remains positive; the separate inclusion using $t - \varepsilon$ above gives the lower bound. Continuity at t, rather than right-continuity alone, is needed on that side.

Source page(s): 10

56 Slutsky's theorem and asymptotic linearity

If $X_n \Rightarrow X$ and $Y_n \xrightarrow{\mathbb{P}} c$ for a constant $c \in \mathbb{R}$, then

$$X_n + Y_n \Rightarrow X + c, \qquad X_n Y_n \Rightarrow cX. \tag{148}$$

For each n the pair X_n, Y_n must be defined together; the limit X need not share their probability space. Independence is not required.

Supplement: The assigned Slutsky proof

For the sum, repeat the preceding cdf bounds comparing $X_n + Y_n$ to $X_n + c$. Choose decreasing positive tolerances for which the shifted arguments are continuity points of the limiting cdf; their complement is countable, so this is possible. Then let the tolerance tend to zero. For the product, $X_n \Rightarrow X$ implies that the sequence is tight: given $\eta > 0$, choose large continuity points $-M, M$ with limiting tail probability below η. Cdf convergence then bounds the tails of X_n for all sufficiently large n. Thus

$$\mathbb{P}(|X_n(Y_n - c)| > \varepsilon) \le \mathbb{P}(|X_n| > M) + \mathbb{P}(|Y_n - c| > \varepsilon/M) \tag{149}$$

tends to at most η, and hence tends to zero after $\eta \downarrow 0$. Since $cX_n \Rightarrow cX$ (by scaling, or trivially for $c = 0$), the sum case applied to $cX_n + X_n(Y_n - c)$ proves the product case. See reference Proposition 8.2.7.

An asymptotically linear estimator has an expansion

$$\widehat{\theta}_n = \theta_0 + \frac{1}{n}\sum_{i=1}^{n} Z_i + R_n, \qquad \sqrt{n} R_n \xrightarrow{\mathbb{P}} 0. \tag{150}$$

If the leading sum satisfies a CLT, for example when Z_i are iid with mean zero and finite variance σ^2, then

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} Z_i + \sqrt{n} R_n \Rightarrow N(0, \sigma^2). \tag{151}$$

Supplement: The assumptions behind the statistical example

The source labels the leading term normal by the CLT without stating assumptions on Z_i. The iid mean-zero finite-variance conditions above are one sufficient set; asymptotic linearity alone does not establish the CLT.

Source page(s): 11–12

57 Lp convergence and bounded convergence in probability

For finite $p \geq 1$, Markov gives

$$\mathbb{P}(|X_n - X| > \varepsilon) \leq \frac{\mathbb{E}|X_n - X|^p}{\varepsilon^p} \longrightarrow 0 \tag{152}$$

when $X_n \to X$ in L^p. For $p = \infty$, essential-supremum convergence gives the same conclusion directly. Conversely, suppose $X_n \xrightarrow{\mathbb{P}} X$ and $|X_n| \leq C < \infty$ a.s. for one common deterministic constant. Then $X_n \to X$ in every finite L^p.

Original TODO: Verify the bound for the limit and finish the truncation argument

For any $a > 0$,

$$\mathbb{P}(|X| > C + a) \leq \mathbb{P}(|X - X_n| > a) \longrightarrow 0. \tag{153}$$

A countable union over $a = 1/m$ gives $|X| \leq C$ a.s. For any $\delta > 0$ split the expectation according to $|X_n - X|^p > \delta$:

$$\mathbb{E}|X_n - X|^p \leq (2C)^p \mathbb{P}(|X_n - X| > \delta^{1/p}) + \delta. \tag{154}$$

Therefore $\limsup_n \mathbb{E}|X_n - X|^p \leq \delta$. Let $\delta \downarrow 0$. Use limsup here because the existence of the expectation limit has not yet been proved (reference Proposition 8.2.9).

The implication diagram in the source is therefore

$$\text{a.s. convergence} \implies \text{convergence in probability} \implies \text{convergence in law}, \tag{155}$$

with L^p convergence also implying convergence in probability. Reverse implications require additional assumptions; convergence in probability always supplies an a.s. convergent subsequence. If a law limit is a constant c, then $X_n \Rightarrow c$ does imply $X_n \xrightarrow{\mathbb{P}} c$, as shown by cdf bounds at $c \pm \varepsilon/2$ (reference Exercise 8.2.6).

Source page(s): 13–14

58 Uniform integrability

A family $\{X_n\}$ is *uniformly integrable* (UI) if

$$\lim_{K \to \infty} \sup_n \mathbb{E}\big[|X_n| \mathbf{1}_{\{|X_n| > K\}}\big] = 0. \tag{156}$$

Equivalently, for every $\varepsilon > 0$ a single cutoff K makes the tail expectation at most ε for every n. A sufficient criterion is the existence of $r > 1$ such that $M_r := \sup_n \mathbb{E}|X_n|^r < \infty$. On $\{|X_n| > K\}$,

$$|X_n| = \frac{|X_n|^r}{|X_n|^{r-1}} \leq \frac{|X_n|^r}{K^{r-1}}, \tag{157}$$

so

$$\sup_n \mathbb{E}\big[|X_n|\mathbf{1}_{\{|X_n|>K\}}\big] \le \frac{M_r}{K^{r-1}} \longrightarrow 0. \tag{158}$$

Original TODO: Why does taking a supremum express uniformity?

The finite number M_r bounds the moment of every variable simultaneously. Consequently the same choice $K > (M_r/\varepsilon)^{1/(r-1)}$ works for all n. Separate finite moments $\mathbb{E}|X_n|^r < \infty$, without a common bound, would not provide that conclusion. This also explains the denominator comparison marked for review on page 14 (reference Proposition 8.3.2).

Source page(s): 15–17

59 Vitali's L1 convergence theorem

If $X_n \xrightarrow{\mathbb{P}} X$ and $\{X_n\}$ is UI, then X is integrable and $\mathbb{E}|X_n - X| \to 0$. In particular this holds under a.s. convergence. The source proves the probability version by truncation; reference Proposition 8.3.1 states the a.s. version. Define the clipping function

$$\phi_K(x) = \max(-K, \min(x, K)). \tag{159}$$

It is continuous, bounded, and $|x - \phi_K(x)| \le |x|\mathbf{1}_{\{|x|>K\}}$.

Supplement: Continuous mapping used in the proof

If $g : \mathbb{R} \to \mathbb{R}$ is continuous and $X_n \xrightarrow{\mathbb{P}} X$, then $g(X_n) \xrightarrow{\mathbb{P}} g(X)$. For a direct argument, restrict to $|X| \le M$ with high probability, use uniform continuity of g on $[-M-1, M+1]$, and bound the remaining error by $\mathbb{P}(|X| > M) + \mathbb{P}(|X_n - X| > \delta)$. Let $n \to \infty$ and then $M \to \infty$. Almost sure convergence is preserved by pointwise continuity. The distribution version follows from the continuous mapping theorem (reference Section 8.7).

By the bounded-convergence-in-probability result, for fixed K,

$$\mathbb{E}|\phi_K(X_n) - \phi_K(X)| \longrightarrow 0. \tag{160}$$

UI gives $\sup_n \mathbb{E}|X_n| < \infty$ (proved below). Take an a.s. convergent subsequence and apply Fatou to obtain $\mathbb{E}|X| < \infty$. Then DCT gives $\mathbb{E}|X - \phi_K(X)| \to 0$. Finally,

$$\mathbb{E}|X_n - X| \le \mathbb{E}|X_n - \phi_K(X_n)| \tag{161}$$

$$+ \mathbb{E}|\phi_K(X_n) - \phi_K(X)| + \mathbb{E}|\phi_K(X) - X|. \tag{162}$$

Original TODO: Make the epsilon argument rigorous; why is X integrable?

Given $\varepsilon > 0$, choose K so the UI tail term and the limit-variable tail term are each below $\varepsilon/3$. Choose N so the bounded middle term is below $\varepsilon/3$ for every $n \ge N$. Their sum is below ε. The integrability of X used for its tail follows from the a.s. subsequence and Fatou, as explained above; it must be established before invoking DCT. For that DCT use $|X - \phi_K(X)| \le |X|$ and pointwise convergence to zero.

Original TODO: Prove the Lp version with the correct UI family

For $1 \le p < \infty$, assume $X_n \xrightarrow{\mathbb{P}} X$ and $\{|X_n|^p\}$ is UI. Fatou along an a.s. subsequence gives $\mathbb{E}|X|^p < \infty$. On the event $A_n = \{|X_n - X| > \delta\}$,

$$\mathbb{E}|X_n - X|^p \le \delta^p + 2^{p-1}\left(\mathbb{E}[|X_n|^p \mathbf{1}_{A_n}] + \mathbb{E}[|X|^p \mathbf{1}_{A_n}]\right). \tag{163}$$

Because $\mathbb{P}(A_n) \to 0$, uniform absolute continuity of UI integrals (proved next) makes the first expectation vanish; integrability of $|X|^p$ makes the second vanish. Let $\delta \downarrow 0$. UI of X_n alone does not imply L^p convergence for $p > 1$. For example $X_n = n^{1/p}\mathbf{1}_{(0,1/n)}$ on $(0,1)$ converges to zero in probability and in L^1, is UI, but $\mathbb{E}|X_n|^p = 1$. No finite-moment UI criterion here asserts convergence in L^∞.

Source page(s): 18–21

60 Equivalent characterization of UI

A family $\{X_n\}$ is UI if and only if both conditions hold:

1. $M := \sup_n \mathbb{E}|X_n| < \infty$.
2. For every $\varepsilon > 0$ there exists $\delta > 0$ such that $\mathbb{P}(A) < \delta$ implies $\sup_n \mathbb{E}[|X_n|\mathbf{1}_A] < \varepsilon$ for every measurable event A.

Proof, forward direction. Choose $a > 0$ with $\sup_n \mathbb{E}[|X_n|\mathbf{1}_{\{|X_n|>a\}}] \le 1$. Then

$$\mathbb{E}|X_n| \le a + 1 \quad \text{for all } n. \tag{164}$$

For any measurable A and any $a > 0$, splitting at a gives

$$\mathbb{E}[|X_n|\mathbf{1}_A] \le a\mathbb{P}(A) + \mathbb{E}[|X_n|\mathbf{1}_{\{|X_n|>a\}}]. \tag{165}$$

Choose a so the second term is uniformly below $\varepsilon/2$ and then $\delta = \varepsilon/(2a)$ to bound the first. **Reverse direction.** Markov gives $\sup_n \mathbb{P}(|X_n| > a) \le M/a$. Choose a large enough that this is less than the δ from condition 2. Apply condition 2 to $A = \{|X_n| > a\}$, for each n, to obtain the UI tail bound (reference Proposition 8.3.3).

Original TODO: Review the implication diagram

A uniformly bounded moment of some order greater than one implies UI; UI implies uniformly bounded first moments. Neither reverse implication holds in general. The two-condition criterion above is the necessary-and-sufficient statement. For instance $X_n = n\mathbf{1}_{(0,1/n)}$ has first moments one but is not UI. A single integrable variable repeated as a constant sequence is UI even if it has no finite moment of any order greater than one.

Source page(s): 19

61 Absolute continuity of an integrable function's integral

For integrable X and every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\mathbb{P}(A) < \delta \quad \Longrightarrow \quad \int_A |X| \, d\mathbb{P} < \varepsilon. \tag{166}$$

By DCT the constant family $X, X, \ldots$ is UI; apply the preceding characterization. Equivalently split the integral at a cutoff and use $\int_A |X| \, d\mathbb{P} \le a\mathbb{P}(A) + \mathbb{E}[|X| \mathbf{1}_{\{|X|>a\}}]$. This is the corollary recorded on page 19 (reference Corollary 8.3.4).

Correction: Small typographical slips in the source and reference

On handwritten page 20 the first equality after splitting the integral must be an inequality when replacing $|X_n|$ by a. The reference's Corollary 8.3.4 ends with $\mathbb{P}(A) < \varepsilon$; its quantified small-probability condition should be $\mathbb{P}(A) < \delta$, as in the statement above. The proof and handwritten version make the intended condition clear.